

Chromatic polynomial

Results:

1. Number of vertices of the graph is the power of highest degree term in λ in the chromatic polynomial.
2. Number of edges in the graph is the absolute value of the coefficient of λ^{n-1} in the chromatic polynomial.
3. Since a bipartite graph is 2-chromatic, $c_2 \neq 0$ in the chromatic polynomial implies that the graph is bipartite.

Model Qn

A graph G has chromatic polynomial $P_G(k) = k^4 - 4k^3 + 5k^2 - 2k$

How many vertices & edges does G have? Is G bipartite?

Justify your answer.

Here no. of vertices = highest

power of $k = 4$

N. of edges = $|\text{coeff. of } k^3| = |-4| = 4 //$

Now to find: ~~the no. of edges~~

~~and whether the graph is bipartite~~

we must expand the chromatic polynomial.

$$P_G(k) = c_1 k + c_2 \frac{k(k-1)}{2!} + c_3 \frac{k(k-1)(k-2)}{3!}$$

$$+ c_4 \frac{k(k-1)(k-2)(k-3)}{4!}$$

G the chromatic polynomial of the graph with 4 vertices.

$$k^4 - 4k^3 + 5k^2 - 2k =$$

$$C_1 k + C_2 \frac{k(k-1)}{2!} + C_3 \frac{k(k-1)(k-2)}{3!}$$

$$+ C_4 \frac{k(k-1)(k-2)(k-3)}{4!}$$

$C_4 = 4!$
equating coeff. of k^3, k^2 , etc
we get the values of C_3, C_2 & so on.

$$\cancel{-4} = \frac{C_3}{3!} + \frac{1 \cdot (k^3 - 3k^2 + 2k)}{(k^2 - 1)(k^2 - 5k + 6)}$$

$$k^4 - 4k^3 + 5k^2 - 2k =$$

$$C_1 k + \frac{C_2 (k^2 - 1)}{2!} + C_3 \frac{(k^3 - 3k^2 + 2k)}{3!}$$

$$+ (k^4 - 5k^3 + 6k^2 - k^3 + 5k^2 - 6k)$$

Coeff. of k^3

$$-4 = \frac{C_3}{3!} + -6 = \frac{C_3 - 36}{6}$$

$$-24 = C_3 - 36 \Rightarrow C_3 = 36 - 24 = \underline{\underline{12}}$$

$$C_3 = \underline{\underline{12}}$$

uQ equating coeff. of k^2 ,

$$5 = \frac{C_2}{2!} - \frac{3C_3}{3!} + 11$$

$$= \frac{C_2}{2} - \frac{3 \times 12}{6} + 11 = \frac{C_2}{2} - 6 + 11$$

$$\frac{C_2}{2} = 5 + 6 - 11 = 0 \Rightarrow \underline{\underline{C_2 = 0}}$$

Here $C_2 = 0$

$\Rightarrow$ the graph G cannot be coloured with 2 colors.

$\therefore G$ is not 2-chromatic

Hence not bipartite.

No. of edges in G is =

$$|\text{coefficient of } k^{n-1}| = |i, \text{ coeff of } k^3|$$

$$= |-4| = \underline{\underline{4}}$$